

Problem 4. Implementing Gradient Computation

$$f\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = x_1 - 2x_2 + \frac{x_3}{4}, \quad g\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + 2x_2 \\ x_2 \\ -x_1 + 3 \end{bmatrix}$$

$$x = \begin{bmatrix} -1 \\ 1 \end{bmatrix} : \text{input to } g$$

$$i) f\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} 1 & -2 & 1/4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + 0 = x_1 - 2x_2 + \frac{x_3}{4}$$

$$ii) g\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} x_1 + 2x_2 \\ x_2 \\ -x_1 + 3 \end{bmatrix}$$

$W_1 \quad (3, 2) \qquad \qquad \qquad b_1 \quad (3, 1)$

$$v) z_2 = f(z_1) = W_2 z_1 + b_2$$

$$\begin{bmatrix} z_1^{(2)} \\ z_2^{(2)} \\ z_3^{(2)} \end{bmatrix} = \begin{bmatrix} W_1^2 & W_2^2 & W_3^2 \\ W_4^2 & W_5^2 & W_6^2 \\ W_7^2 & W_8^2 & W_9^2 \end{bmatrix} \begin{bmatrix} z_1^{(1)} \\ z_2^{(1)} \\ z_3^{(1)} \end{bmatrix} + \begin{bmatrix} b_1^2 \\ b_2^2 \\ b_3^2 \end{bmatrix}$$

$$\frac{\delta z_2}{\text{svec}(W_2)} = \begin{bmatrix} \frac{\delta z_2}{\delta W_1^2} & \frac{\delta z_2}{\delta W_2^2} & \frac{\delta z_2}{\delta W_3^2} & \frac{\delta z_2}{\delta W_4^2} & \frac{\delta z_2}{\delta W_5^2} & \frac{\delta z_2}{\delta W_6^2} & \frac{\delta z_2}{\delta W_7^2} & \frac{\delta z_2}{\delta W_8^2} & \frac{\delta z_2}{\delta W_9^2} \\ \frac{\delta z_2}{\delta W_1^2} & \frac{\delta z_2}{\delta W_2^2} & \frac{\delta z_2}{\delta W_3^2} & \frac{\delta z_2}{\delta W_4^2} & \frac{\delta z_2}{\delta W_5^2} & \frac{\delta z_2}{\delta W_6^2} & \frac{\delta z_2}{\delta W_7^2} & \frac{\delta z_2}{\delta W_8^2} & \frac{\delta z_2}{\delta W_9^2} \\ \frac{\delta z_2}{\delta W_1^2} & \frac{\delta z_2}{\delta W_2^2} & \frac{\delta z_2}{\delta W_3^2} & \frac{\delta z_2}{\delta W_4^2} & \frac{\delta z_2}{\delta W_5^2} & \frac{\delta z_2}{\delta W_6^2} & \frac{\delta z_2}{\delta W_7^2} & \frac{\delta z_2}{\delta W_8^2} & \frac{\delta z_2}{\delta W_9^2} \end{bmatrix} = \begin{bmatrix} z_1' & z_2' & z_3' & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & z_1' & z_2' & z_3' & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & z_1' & z_2' & z_3' \end{bmatrix}$$

$\frac{\delta z_2}{\delta b_2} = 1$

$$\text{vi)} \quad z_1 = g(x) = W_1 x + b_1$$

$$f(z_1) = W_2 z_1 + b_2$$

$$\frac{\delta(f \circ g)}{\delta z_1}(x) = W_2$$

$$\text{vii)} \quad y = (f \circ g)(x), \quad z = g(x) = W_1 x + b_1$$

$$y = f(z_1) = W_2 z_1 + b_2$$

$$= W_2(W_1 x + b_1) + b_2$$

$$\frac{\delta(f \circ g)}{\delta b_1} = \frac{\delta y}{\delta b_1} = \frac{\delta y}{\delta z_1} \frac{\delta z_1}{\delta b_1} = p^T \frac{\delta z_1}{\delta b_1} = p^T$$

$$\frac{\delta(f \circ g)}{\delta W_1} = \frac{\delta y}{\delta W_1} = \text{unvec} \left[\frac{\delta y}{\delta z_1} \frac{\delta z_1}{\text{vec}(W_1)} \right] = p x^T$$